

Future Mobility Maker: A Conversational AI Experience for Public Exhibitions

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Figure 1: Overview of the *Future Mobility Maker* experience. Visitors engage in spoken dialogue with life-sized AI agent Saya, who elicits their interests and instantly generates a personalized future mobility vision, visualized and explained in real time.

Abstract

Large-scale exhibitions rarely allow visitors to actively relate future visions to their own lives. We present *Future Mobility Maker*, a visitor-participatory demo in which exhibition visitors converse

with a life-sized AI agent to instantly generate personalized visions of future mobility. Through a short multi-turn dialogue, the agent elicits visitors' everyday interests and sensory impressions, and transforms their spoken narratives into a unique future mobility image accompanied by an in-character explanation. This interaction enables visitors to experience future mobility as a personally constructed concept rather than a fixed exhibit. The system employs face-based automatic dialogue initiation and a short, self-contained interaction flow, enabling operation in high-traffic public exhibition environments without staff intervention. We invite CHI



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participants to experience *Future Mobility Maker* and reflect on how conversational generative AI can support personalized co-creative experiences in public spaces.

CCS Concepts

• **Human-centered computing** → **Displays and imagers.**

Keywords

Conversational AI, Co-Creation, Public Exhibitions, Generative AI, AI agent

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1 Introduction & Background

Large-scale public exhibitions play an important role in presenting emerging technologies and future visions of society. Visitors encounter ideas about future housing, mobility, energy, and lifestyles, often at a scale and level of realism that cannot be experienced elsewhere. However, despite addressing topics that are closely tied to everyday life and personal values, exhibitions often foreground expert-authored narratives, leaving limited room for visitors to actively relate these visions to their own experiences, interests, or aspirations [13]. Recent HCI work has therefore explored exhibition formats that move beyond one-way presentation by explicitly inviting public participation in ideation and reflection [3, 4, 13].

In parallel, conversational agents and generative AI have attracted growing attention as technologies that can support interactive and personalized experiences [11]. Prior work has explored conversational guide systems and information agents deployed in public spaces, including museum guides [2, 7], receptionist-style information agents in buildings [9], and robots deployed in transit hubs [5, 12] and shopping environments [6]. Separately, generative AI has been explored in public exhibitions [8, 10] and cultural/educational experiences [14] to transform participant input into personalized media artifacts and interactive narratives. However, many exhibition deployments still rely on constrained inputs (e.g., menu-like choices), limiting visitors' ability to express personal narratives through open-ended speech.

In the context of public exhibitions, spoken conversation can offer a promising modality for participatory future-making experiences. Open-ended dialogue allows visitors to express everyday interests, memories, and sensory impressions in their own words, and participatory speculative design in public settings shows how such situated narratives can ground collective imagination about alternative futures [3, 4]. By positioning generative AI not as an automatic content generator but as a conversational partner that listens and responds, exhibitions can shift from spaces of presentation to spaces of co-creation, where future visions emerge through interaction [2, 11].

In this demonstration, we present *Future Mobility Maker*, a visitor-participatory co-creation experience in which exhibition visitors

engage in short, multi-turn spoken dialogue with a life-sized, photorealistic AI agent. During the interaction, the agent elicits visitors' interests, desired destinations, and sensory impressions, and immediately transforms these personal narratives into a unique future mobility image accompanied by an in-character explanation. Rather than presenting future mobility as a predefined concept, the system allows visitors to experience it as something constructed from their own words. By presenting this experience as an interactive, walk-up demo, we invite CHI participants to reflect on how conversational generative AI can enable personalized, engaging, and socially situated co-creative experiences.

2 Future Mobility Maker

We present *Future Mobility Maker*, an exhibition system in which the photorealistic conversational agent Saya [1] engages visitors in several turns of dialogue. Based on visitors' preferences and interests elicited through conversation, the system generates personalized future mobility images using DALL-E3¹ and presents them with explanations.

2.1 Physical Environment Configuration

This system consists of a desktop PC connected to a microphone (Fairy Devices T-03C), speaker (Sanwa Supply 400-SP088), camera (Logicool BRIO C1000eR), and a 65-inch 4K display (Panasonic TH-65EQ2J) for life-size, high-definition rendering of Saya. We designed the enclosure to conceal the microphone, camera, and speaker from view, aiming to create an environment where visitors can engage in natural dialogue without being conscious of the devices.

As shown in Fig. 2, the system comprises three main modules—recognition, dialogue management, and output generation—that coordinate for real-time response generation and dialogue progression.

2.2 Recognition

This system incorporates multimodal recognition capabilities that process image and audio inputs in real time.

Image recognition. The system determines user presence through face detection and uses this as a trigger for dialogue initiation and termination. For detected users, age estimation and position information enable child detection, activating a child-responsive mode where Saya adopts a crouching posture for eye-level contact and uses informal, child-friendly language. Mouth shape analysis extracts speech segments, selectively accepting only the voice of the user directly in front. User change detection resets dialogues, maintaining smooth operation in high-turnover environments.

Speech recognition. The system extracts speech segments through volume-based noise gate processing, Silero VAD (Voice Activity Detection), and image-based utterance detection, then performs speech recognition using YY System² (multilingual speech recognition). Results feed into dialogue flow control and output generation.

2.3 Dialogue Management

Dialogue flow control determines actions based on a predefined scenario: dialogue initiation, four questions, image generation, image

¹<https://openai.com/dall-e>

²<https://yysystem.com>

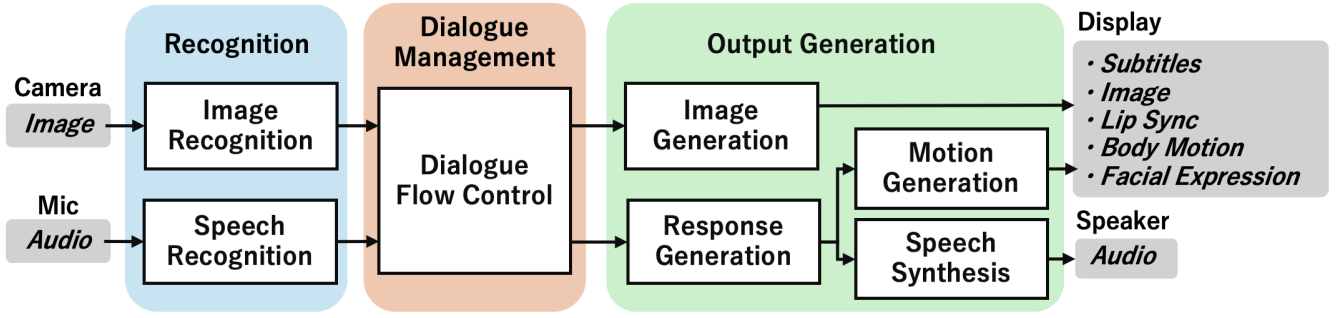


Figure 2: System overview. The *Future Mobility Maker* system comprises seven components (image recognition, speech recognition, dialogue flow control, image generation, response generation, motion generation, and speech synthesis).

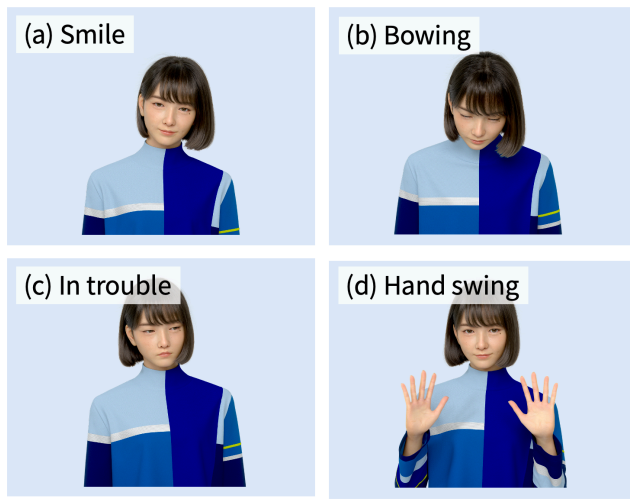


Figure 3: Examples of Saya's automatically generated facial expressions and gestures synchronized with her spoken responses. Nonverbal behaviors are assigned at the utterance level based on semantic tags (e.g., greeting, agreement, clarification), supporting natural and expressive dialogue during the interaction.

explanation, and dialogue termination (see Fig. 1). First, the system initiates dialogue triggered by user face detection and presents the four questions sequentially. At this point, GPT-4o-mini judges whether each user response to each question is appropriate, and if insufficient, the system provides answer hints to help the user respond. Turn-taking integrates the processes and results of speech segment extraction, speech recognition, appropriate timing judgment for turn exchange (TRP; Transition Relevance Place), and output generation to detect the end of user utterances and initiate responses at appropriate timing. If face detection is interrupted for 2 seconds, the dialogue terminates regardless of scenario progress, enabling flexible user-state-responsive control.

2.4 Output Generation

The output generation module produces Saya's responses and generates corresponding motions and synthesized speech, while also generating images in real time.

Response generation. The system inputs selected prompts, past speech recognition results, and Saya's previous utterances into GPT-4o-mini to generate context-appropriate, character-consistent utterances. For privacy, audio and text data are not used for model training; audio is not stored, and text is stored locally only for operational monitoring without user linkage.

The system generates Saya's body motions and facial expressions based on semantic tags attached to utterances (Fig. 3). Seven tag types are assigned per sentence: [0] Neutral, [1] Agreement, [2] Disagreement, [3] Brief backchannel, [4] Question, [5] Greeting, [6] Farewell. For example, a properly formatted output follows the pattern:

"I'm Saya, an AI designer. [5] Let's design the future mobility that you envision together. [0]"

This structure assigns nonverbal expressions per sentence.

The system uses Azure Speech Studio to generate clear audio while preserving Saya's voice quality. Lip sync calculates mouth shape transitions corresponding to phonemes based on the generated utterance content and synchronizes them with speech synthesis.

Image generation. The system integrates visitor preferences with base prompts to generate future mobility images via DALL-E3. Fig. 4 shows example images generated from the conversation.

Subsequently, the system renders an image on the left portion of the display according to generated facial expressions, body motions, and scenarios, and plays synthesized speech at appropriate timing. Seamless motion transitions are achieved through real-time connection of locally pre-rendered 4K video segments, displaying Saya's expressions and gestures in fine detail. For audio output, the system incorporates volume adjustment considering the acoustic environment of the event venue, providing a natural and comfortable listening experience for visitors. Additionally, by displaying visitor utterance content from speech recognition and Saya's responses as subtitles on the right portion of the display, the system adds visual assistance to achieve more accurate and smooth communication.



Figure 4: Examples of future mobility generated through dialogue with visitors. Original images generated by DALL-E3 based on visitor responses (favorite activities, desired destinations, sounds they would hear, and atmosphere they want to create).

3 Interactive Demo Description

In the CHI interactive demo, participants will experience *Future Mobility Maker* as an interactive, walk-up exhibition system designed for high-traffic public spaces. The demo is designed to support short, self-contained experiences that can be enjoyed individually or observed by surrounding participants, making it suitable for conference environments.

3.1 Walk-up Interaction and Dialogue Flow

The demo begins when a participant stands in front of the life-sized AI agent, Saya. Using face detection, the system automatically initiates the interaction without requiring explicit input. Saya introduces herself and briefly explains that she will help the participant design their own future mobility through conversation.

The interaction consists of a short multi-turn dialogue structured around four questions. These questions are designed to gradually elicit personal and imaginative content, starting from familiar everyday topics and progressing toward more abstract future-oriented concepts. Specifically, Saya asks about (1) the participant’s interests, (2) places they would like to go, (3) sounds associated with that place, and (4) the atmosphere they would like to create. Throughout the dialogue, Saya responds with brief backchannels and follow-up prompts, encouraging participants to continue speaking while maintaining a natural conversational rhythm.

If a participant’s response is brief or unclear, Saya provides gentle hints and re-asks the question, supporting dialogue continuation without rigid constraints. If the participant steps away, the interaction terminates and the system resets for the next visitor.

3.2 Co-Creation and Visualization Experience

After the question phase, Saya informs the participant that she will generate a future mobility concept based on their responses. During this brief waiting period, the system uses generative AI to transform the participant’s spoken narratives into a personalized future mobility image.

Once generation is complete, the image is displayed alongside Saya, who presents the result with an in-character explanation. The explanation incorporates elements from the participant’s earlier responses, such as interests, destinations, and sensory impressions, reinforcing the sense that the outcome was constructed from

the participant’s own words. Participants can react by verbally commenting on the generated mobility, comparing it to their expectations, or sharing interpretations with companions or nearby observers.

To extend the experience beyond the immediate interaction, participants are invited to download their generated image via a QR code. This allows them to take the result with them and revisit the experience after leaving the demo space.

4 Future Directions

Several directions emerge for future exploration of this interaction approach. While the current demonstration focuses on a single-pass co-creation experience, future systems could support more iterative or interactive forms of dialogue, allowing visitors to further shape or revise generated outcomes. Such iterative capabilities would enable deeper personalization and greater visitor agency in the co-creation process.

In addition, the underlying interaction approach—eliciting personal narratives through conversation and immediately visualizing them—could be extended beyond future mobility to other exhibition and public engagement contexts. These directions point toward broader opportunities for conversational generative AI to support participatory, experience-centered design in public spaces.

5 Conclusion

We presented *Future Mobility Maker*, a visitor-participatory demonstration using conversational generative AI to enable personalized co-creation in public exhibition spaces. By engaging visitors in short, spoken dialogue with a life-sized AI agent and immediately transforming their narratives into visualized future mobility concepts, the system reimagines exhibitions as spaces of interaction rather than passive presentation.

Through this demo, we aim to provoke reflection on how conversational AI and generative technologies can support meaningful, personalized engagement at scale, particularly in environments characterized by brief encounters and high visitor turnover. By foregrounding experience over evaluation, *Future Mobility Maker* invites the CHI community to explore new possibilities for designing co-creative, socially situated interactions that connect emerging technologies with people’s everyday lives.

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